Introduction to Software Systems (COMP 206)
Midterm 1 – Version 2

Instructor: Mona Elsaadawy and Jacob Errington

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- This exam is printed on both sides.
- A 8.5x11 inch crib sheet, handwritten only on one side, is permitted.
- Calculators are not permitted.
- All answers must be written directly on the exam paper.

Question	Points	Score
Multiple-choice questions	13	
Short answers, big ideas	7	
LogStats Pro	10	
Total:	30	

Best of luck! – Mona & Jake

- "Select all that apply" questions award partial marks. Each correct choice gives 0.5 points.
- Select your answer(s) by clearly circling the letter(s) of the choice.

Question 1: Multiple-choice questions (13 points)

1. (2 points) Which of the following are responsibilities of an operating system? (Select all that apply.)
   (a) Manage the computer's resources.
   (b) Provide applications to users such as a web browser.
   (c) Convert program source code into machine code to run on the CPU.
   (d) Provide libraries to application developers.

2. (2 points) Which of the following are true about the Linux kernel? (Select all that apply.)
   (a) It is an operating system.
   (b) It is inspired by Unix.
   (c) You can download its source code and make modifications to it.
   (d) It is only used on servers.

3. (2 points) Which of the following commands **definitely** involve an **absolute path**? Assume that standard environment variables have their usual values, but make no assumptions about unknown environment variables. (Select all that apply.)
   (a) `cp $FILE $FILE2`
   (b) `cat ~/.bashrc`
   (c) `cp $BACKUP_DIR/2025-09-12-notes.md notes`
   (d) `tar czvf backups/2025-10-01.tar.gz $HOME`

4. (2 points) Consider the following output of `ls -a`

   ```
   .?report.pdf
   .report.pdf
   report.pdf
   report.pdf.gz
   ```

   Suppose that all of those entries are *files* and that we have write permission on the current directory. Suppose that we run `rm .?*.pdf`

   Which files will be deleted? (Select all that apply.)
   (a) `.?report.pdf`
   (b) `.report.pdf`
   (c) `report.pdf`
   (d) `report.pdf.gz`

5. (1 point) Which of the following permissions expressed in octal notation correspond to the symbolic permissions string rw-r--r--?
   (a) 655
   (b) 322
   (c) 311
   (d) 644

6. (2 points) Consider the octal permissions 631 on a file owned by user alice and group users. Suppose that user bob belongs to the group users, but that the user claire does not belong to the group users. Which of the following statements are true? (Select all that apply.)
   (a) alice can read the file.
   (b) alice can execute to the file.
   (c) bob can execute the file.
   (d) claire can read the file.

7. (2 points) In which of the following commands is `grep` reading from the standard input? (Select all that apply.)

   (a) `$ grep -i 'au' log.txt`
   (b) `$ grep 'error' < system.log`
   (c) `$ cat list.txt | grep -n 'rd'`
   (d) `$ grep "bash" $(which config)`
   Hint: The which command shows the path of the executable file that will run when you type a command— in this example case, config)

Question 2: Short answers, big ideas (7 points)

In this section, every time you give a command as an answer to a question, you should assume that the command's effect is performed. It's as if you were actually running these commands yourself. Pay attention to what directory you're in as you go, and write accordingly correct paths.

You sit down at your workstation, ready to start the day working for BigIfTrue Inc., a small, big-data company.

1. (1 point) Your username is dev206 which is your employee ID number. The server you do your work on is called devserver.bigiftrue.com.

   Give a command to log into that server remotely.

   _____

2. (1 point) Your manager places a file called todo.txt into your home directory every day with some tasks for you to do.

   Give a command that prints out the first five lines of that file on the terminal.

   _____

   The output of the command is this:

   Your task for today is to build a bash prototype of our LogStats Pro platform.
   There's some starter code for you to build on in: /shared/LWpro

3. (1 point) After a week of working at BigIfTrue, you're annoyed by manually typing a command each time you log in to view the todo list.

   Give a command that appends `cat ~/todo.txt` to the end of a bash configuration file that is read each time you login (Assuming that your default shell is Bash)

   _____

4. (1 point) You decide to copy the entire starter code directory into the `projects` directory that already exists in your home.

   Give a command to perform that copy.

   _____

5. (1 point) You enter the `projects` directory, and decide to create an archive of the starter code directory before getting started.

   Give a tar command to create a compressed archive under `~/backups`, with a name of your choosing, of the copy you made of the starter code directory.

   _____

6. (2 points) You enter the starter code directory. Inside, you find a README file that contains the following.

   We are investigating a stability issue with our web application. The application logs are stored, per service and date, under:

   /var/log/webapp/DATE.log

   where DATE look like 2025-03-20.

   You decide to look at the first five rows of /var/log/webapp/ 2025-03-20.log, which contains:

   LEVEL, SERVICE, MESSAGE, TIMESTAMP
   ERROR, api, Database connection timeout, 2025-03-20T10:15:32Z
   INFO, api, User login successful, 2025-03-20T10:15:33Z
   ERROR, database, Connection pool exhausted, 2025-03-20T10:16:05Z
   WARN, payment_gateway, High response time, 2025-03-21T09:10:01Z

   Give a command, involving uses of pipes, to print only the timestamps of the ERROR messages in that file.

   _____

Question 3: LogStats Pro

(10 points)

Now that you've explored the situation in the previous section, you're ready to develop a more complete prototype of the LogStats Pro system in a Bash script called Logstats.

In a check-in with your manager, she explains that the system must identify which services are generating the most errors. The script should analyze multiple log files, **similar in format to the one shown in the final short-answer question**, and rank the services by error frequency.

The script should take one argument: a **path to a file** that itself will contain a list of (absolute) paths to various log files (one per line).

Your script should analyze all the log files in the list and print a summary. It should print the number of ERROR log messages for each service, sorted in descending order (service with the most error messages first).

The script will be called as: ./logstats FILELIST

For example: when called as `./logstats /tmp/my_logs.txt` and `my_logs.txt` may contain the following:

```
/var/log/webapp/2025-03-20.log
/var/log/webapp/2025-03-21.log
/var/log/webapp/2025-03-22.log
```

Your script might output the following:

```
15 api
8  database
3 payment_gateway
```

Evaluation:

- You get 6 points for basic functionality: reading at least one line from the given file, filtering for error log messages, counting the occurrences per service, and printing the services in descending order of frequency.
- You get 2 points when your script properly aggregates across *all* files given in the list.
- You get one point for validating the inputs of the script: there should be exactly one command- line arguments, and the given file should refer to a file that exists.
- You get one point for appropriately printing a warning message when any of the files in the list doesn't exist.
- You may lose one (or more) points when the code is excessively complicated or difficult to read.
- You will lose a point for the use of temporary files – they aren't necessary to solve this problem and result in a suboptimal solution.

You may use the rest of this page to write down ideas and a draft implementation.

Write your final answer on the following page. Only that page will be graded as an answer to this question.

Write your solution to the long-answer question on this page.

Scratch space